CLASS XI CHEMISTRY — CHAPTER 2

STRUCTURE OF ATOM

FULL DETAILED NOTES • JEE + SCHOOL • CONCEPTS + EXAMPLES + FORMULAS

1. SUB-ATOMIC PARTICLES

J. J. Thomson's cathode-ray experiments established the electron. Canal/positive rays led to the identification of positive particles, and the proton is the hydrogen nucleus. James Chadwick discovered the neutron.

Electron: charge $-1.602\times10^{\blacksquare1\blacksquare}$ C; mass $9.109\times10^{\blacksquare31}$ kg. **Proton:** $+1.602\times10^{\blacksquare1\blacksquare}$ C; mass $1.673\times10^{\blacksquare2\blacksquare}$ kg. **Neutron:** charge 0; mass $1.675\times10^{\blacksquare2\blacksquare}$ kg.

2. THOMSON MODEL

Atom was proposed as a positively charged sphere with electrons embedded in it. It explained overall electrical neutrality but failed to explain alpha-particle scattering and the nuclear structure.

3. RUTHERFORD MODEL ■■■

Alpha particles were fired at thin gold foil. Most passed straight through, some deflected, and a tiny fraction underwent large-angle scattering.

Conclusions: most of the atom is empty space; nearly all positive charge and mass are in a tiny dense nucleus; electrons occupy the surrounding region. **Limitation:** classical physics could not explain atomic stability or line spectra.

4. ELECTROMAGNETIC RADIATION ■■■

Wavelength λ is the distance between successive wave crests; frequency ν is cycles per second. In vacuum, $c\approx3.00\times10^{\blacksquare}$ m $s^{\blacksquare1}$ and **$c=\lambda\nu$**. Photon energy is **$E=h\nu=hc/\lambda$**, $h=6.626\times10^{\blacksquare3\blacksquare}$ J s.

Example: $\lambda=600$ nm $\rightarrow$ $\nu=(3\times10^{\blacksquare})/(600\times10^{\blacksquare\blacksquare})=5\times10^{1\blacksquare}$ Hz.

5. ELECTROMAGNETIC SPECTRUM

Increasing frequency/energy: radio → microwave → infrared → visible → ultraviolet → X-rays → gamma rays. Increasing frequency means decreasing wavelength and increasing photon energy.

6. PLANCK QUANTUM THEORY ■■■

Energy is exchanged in discrete packets (quanta). One photon has energy $E=h\nu$. **Example:** $\nu=5\times10^{1\blacksquare}$ Hz gives $E\approx3.313\times10^{\blacksquare1\blacksquare}$ J.

7. PHOTOELECTRIC EFFECT ■■■

Light above a threshold frequency $\nu_{\blacksquare}$ can eject electrons from a metal. Einstein equation: **$h\nu=h\nu_{\blacksquare}+K.E._{max}$**. Thus $K.E._{max}=h(\nu-\nu_{\blacksquare})$. Frequency controls maximum kinetic energy; above threshold, intensity mainly affects the number of emitted electrons.

8. ATOMIC SPECTRA

Continuous spectra contain a continuous range of wavelengths. Line spectra contain discrete wavelengths and show that atomic energy levels are quantized.

9. BOHR MODEL ■■■

For hydrogen-like atoms, electrons occupy stationary allowed orbits. Angular momentum is quantized: **$mvr=nh/2\pi$**. Radiation is emitted/absorbed only when an electron changes level, with photon energy equal to the energy difference.

Energy: E■ $=-13.6Z^2/n^2$ eV $=-2.18\times10$■¹■ Z^2/n^2 J.

Example: H at n=2: E■=–3.40 eV. He■ at n=2: E■=–13.6 eV.

10. BOHR RADIUS ■■■

r■ $=0.529\ n^2/Z$ Å. Radius increases as n^2 and decreases as nuclear charge Z increases.

Example: H, n=2 → r=2.116 Å; He■, n=2 → r=1.058 Å.

11. HYDROGEN SPECTRAL SERIES ■■■

Rydberg equation: **$1/\lambda=RZ^2$(1/n■²–1/n■²)**, n■>n■. Lyman n■=1 (UV); Balmer n■=2 (visible/near UV); Paschen n■=3 (IR); Brackett n■=4 (IR); Pfund n■=5 (IR).

Example: n=3→2 is Balmer; n=2→1 is Lyman.

12. LIMITATIONS OF BOHR MODEL

It works well for one-electron species but does not fully explain multi-electron spectra, fine structure, Zeeman/Stark effects, or the wave nature and uncertainty of electrons.

13. DUAL NATURE OF MATTER ■■■

de Broglie relation: **$\lambda=h/p=h/mv$**. For kinetic energy K, **$\lambda=h/\sqrt{2mK}$**. For an electron accelerated through V (non-relativistic), **$\lambda=h/\sqrt{2meV}$**.

Example: At the same momentum, particles have the same de Broglie wavelength; at the same speed, the lighter particle has the larger wavelength.

14. HEISENBERG UNCERTAINTY PRINCIPLE ■■■

$\Delta x\Delta p\geq h/4\pi$. Position and momentum cannot both be known with unlimited precision. This supports the probability/orbital description rather than fixed classical paths.

15. QUANTUM MECHANICAL MODEL

An orbital is a region of high probability of finding an electron. Electrons are described using wave functions and quantum numbers; an orbital is not a circular path.

16. FOUR QUANTUM NUMBERS ■■■

Number	Symbol	Allowed values	Meaning
Principal	n	1,2,3,…	Shell, size, main energy
Azimuthal	l	0 to n–1	Subshell and shape
Magnetic	m■	–l to +l	Orbital orientation
Spin	m■	±1/2	Spin state

17. SHELL, SUBSHELL & ORBITAL CAPACITY ■■■

Shell n has maximum **$2n^2$** electrons and n^2 orbitals. A subshell with quantum number l has **2l+1** orbitals and maximum **2(2l+1)** electrons. s: 1 orbital/2e■; p: 3/6; d: 5/10; f: 7/14.

Example: n=3 has 9 orbitals and maximum 18 electrons. A 3p subshell has l=1, m■=–1,0,+1, so it has 3 orbitals and 6-electron capacity.

18. ORBITAL SHAPES & NODES ■■

s orbitals are spherical; p orbitals are dumbbell-shaped; d and f orbitals have more complex shapes. Total nodes=n–1; angular nodes=l; radial nodes=n–l–1.

Example: 2p has 1 total node, 1 angular and 0 radial. 3s has 2 total, 0 angular and 2 radial nodes.

19. ELECTRONIC CONFIGURATION ■■■

Electrons fill orbitals using Aufbau principle, Pauli exclusion principle and Hund's rule. Common order: 1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p, 6s, 4f, 5d, 6p, 7s.

Pauli: no two electrons have the same four quantum numbers; one orbital holds two opposite-spin electrons. **Hund:** degenerate orbitals fill singly with parallel spins before pairing.

20. CONFIGURATION EXAMPLES ■■■

Na (11): $1s^2 2s^2 2p$■$3s^1$=[Ne]$3s^1$. **Cl (17):** [Ne]$3s^2 3p$■. **Ca (20):** [Ar]$4s^2$. **Fe (26):** [Ar]3d■$4s^2$.

21. Cr & Cu EXCEPTIONS ■■■

Cr (Z=24) = **[Ar]3d■$4s^1$**; Cu (Z=29) = **[Ar]$3d^1$■$4s^1$**. Half-filled d■ and fully filled d^1■ arrangements gain extra stability from exchange/symmetry effects.

22. TRANSITION-METAL IONS ■■■

When transition-metal cations form, 4s electrons are generally removed before 3d electrons.

Example: Fe=[Ar]3d■$4s^2$; Fe^2■=[Ar]3d■; Fe^3■=[Ar]3d■.

23. ISOTOPES, ISOBARS & ISOTONES ■■

Isotopes: same Z, different A (^{1}H,^{2}H,^{3}H). **Isobars:** same A, different Z (■■Ar,■■Ca). **Isotones:** same neutron number. **A=Z+N**.

24. AVERAGE ATOMIC MASS ■■■

Average atomic mass = Σ(isotopic mass × fractional abundance).

Example: masses 10 and 11 with 20% and 80% abundance: average=10(0.20)+11(0.80)=10.8 u.

25. HIGH-YIELD FORMULA SHEET ■■■

$c=\lambda\nu$; $E=h\nu=hc/\lambda$; K.E.max=$h(\nu-\nu_\blacksquare)$; $mvr=nh/2\pi$; $E_\blacksquare=-13.6Z^2/n^2$ eV; $r_\blacksquare=0.529n^2/Z$ Å; $1/\lambda=RZ^2(1/n_\blacksquare^2-1/n_\blacksquare^2)$; $\lambda=h/mv$; $\lambda=h/\sqrt{(2mK)}$; $\Delta x\Delta p\geq h/4\pi$; shell capacity=$2n^2$; shell orbitals=$n^2$; subshell orbitals=2l+1; total nodes=n–1; angular nodes=l; radial nodes=n–l–1.

26. COMMON JEE TRAPS ■■■

• Bohr formulas directly apply to hydrogen-like one-electron species. • 4s fills before 3d in the common Aufbau order, but 4s electrons are removed first from transition-metal cations. • Photon energy depends on frequency, not intensity. • An orbital is a probability region, not a path. • For 3d, n=3 and l=2. • Always convert nm/Å to SI units before numerical calculation.

27. SCHOOL + JEE REVISION CHECKLIST

✓ Subatomic particles and experiments ✓ atomic models and limitations ✓ electromagnetic radiation ✓ Planck/photoelectric effect ✓ spectra ✓ Bohr model ✓ Rydberg series ✓ de Broglie ✓ uncertainty ✓ quantum mechanical model ✓ four quantum numbers ✓ orbital shapes/nodes ✓ Aufbau/Pauli/Hund ✓ Cr/Cu ✓ ions ✓ isotopes/isobars/isotones ✓ average atomic mass ✓ numerical practice.

CONCEPT → EXAMPLE → PRACTICE → JEE + SCHOOL SCORE